



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO JAI MARUTHI TRADERS

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

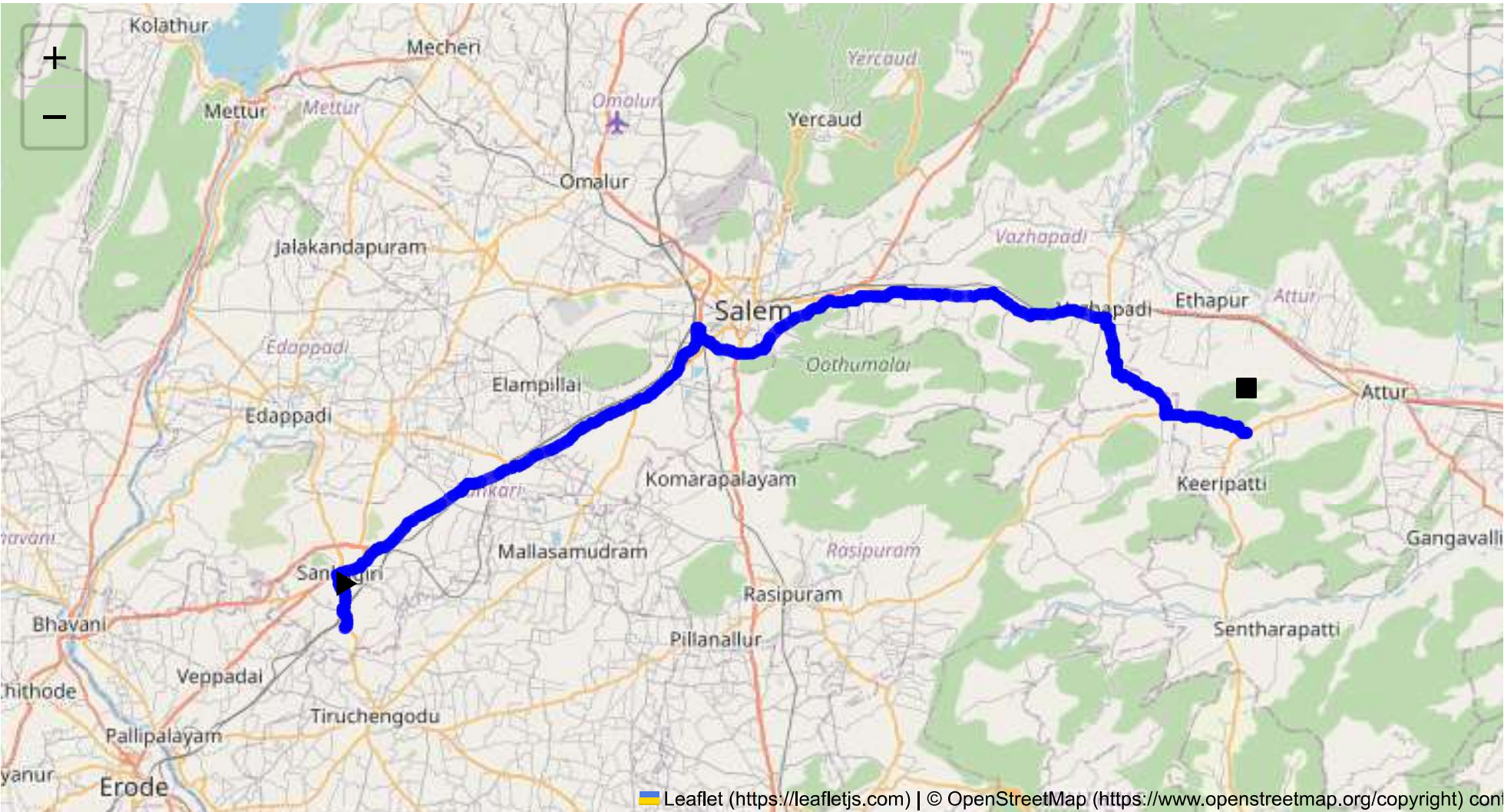
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 89.41 km
Estimated Duration: 1.8 hours
Adjusted Duration (Heavy Vehicle): 2.3 hours
Start: (11.4381, 77.8734)
End: (11.571136, 78.498505)



Welcome to the Journey Risk Management Study

Route Safety Analysis: From CVQF+23W, Sangagiri to HFCX+HCW, Malliakkarai, Tamil Nadu

1. Overview of the Route Map: The route primarily follows major highways and state roads, including SH 79, which is crucial for local and through traffic. The journey is approximately 89.41 kilometers and typically takes around 1.84 hours under normal conditions. The roads are generally well-marked, with signage indicating directions and key turnoffs.

2. Weather Conditions: Typical weather includes a tropical climate with high temperatures and monsoon seasons, potentially affecting road conditions. Developments such as heavy rain could cause slippery roads, reduced visibility, and flooding, particularly during the South-West (June to September) and North-East monsoon periods (October to December).

3. Traffic Patterns: Traffic congestion is likely during peak hours, typically 8-10 AM and 5-7 PM. Local markets and town centers, especially near Salem and other rural junctions, can experience slower movement of heavy vehicles. SH 79 may have stretches prone to congestion due to urban and semi-urban sprawl.

4. Road Quality and Infrastructure: The roads vary in condition: highways are generally well-maintained but may have sporadic potholes and ongoing construction activity in certain stretches. Rural roads might be narrower, requiring caution for heavy vehicles, especially near populated or agricultural areas.

5. Alternative Routes: For emergencies or severe congestion, alternative routes using NH 544 towards Salem and then branching to reach Malliakkarai can be considered. This may extend travel times but provides more infrastructure and service facilities.

6. Local Regulations: Regulations stipulate specific travel times for heavy vehicles carrying hazardous materials, often restricted to non-peak hours to minimize risks. Compliance with state transportation laws and safety standards for hazardous materials is essential.

7. Historical Incidents: There have been reports of accidents involving heavy vehicles in the region predominantly related to challenging monsoon conditions and vehicle instability on curvy road sections. Ensuring proper vehicle maintenance and driver awareness training is vital.

8. Environmental Considerations: The route passes near agricultural lands and small water bodies, requiring strict adherence to no-spill protocols to prevent environmental damage. Low emission standards and noise control around populated areas are also mandated.

9. Communication Coverage: Mobile network coverage is generally reliable in urban and suburban areas. However, there may be potential dead zones in more rural segments, requiring pre-planned communication checks and GPS-based navigation backups.

10. Emergency Response Times: Response times in urban sections, like those near Salem, could be faster (30 minutes to 1 hour), whereas service in rural regions might take longer, potentially over an hour due to limited accessibility for emergency vehicles.

12. Overall Summary of Risk Assessment: The route poses moderate risks mostly related to weather and road conditions, with occasional congestion. Proper planning, compliance with local regulations, and awareness of alternative routes can mitigate potential delays and hazards. Training drivers on local environmental considerations and ensuring communication devices are fully operational before departure can further enhance safety.

Conclusion: This assessment highlights the necessity for thorough preparation when transporting hazardous materials. Emphasizing driver training, adherence to safety protocols, and maintaining communication readiness will ensure smoother and safer transportation.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.63303, 78.12713	15 KM/Hr
1	U-Turn	High	11.5709952, 78.4994313	10 KM/Hr
2	Turn	High	11.43968, 77.87345	15 KM/Hr
3	Turn	High	11.44029, 77.87544	15 KM/Hr
4	Turn	Medium	11.44900, 77.87406	30 KM/Hr
5	Turn	Medium	11.45000, 77.87213	30 KM/Hr
6	Turn	High	11.45028, 77.87202	15 KM/Hr
7	Turn	Medium	11.45309, 77.87325	30 KM/Hr
8	Turn	Medium	11.46400, 77.87323	30 KM/Hr
9	Turn	Medium	11.46765, 77.87037	30 KM/Hr
10	Turn	High	11.47403, 77.86841	15 KM/Hr
11	Turn	High	11.49149, 77.89601	15 KM/Hr
12	Turn	High	11.49224, 77.89601	15 KM/Hr
13	Turn	Medium	11.64172, 78.11817	30 KM/Hr
14	Turn	Medium	11.64216, 78.11875	30 KM/Hr
15	Turn	Medium	11.64195, 78.12129	30 KM/Hr
16	Turn	Medium	11.64162, 78.12160	30 KM/Hr
17	Turn	Medium	11.64120, 78.12168	30 KM/Hr
18	Turn	Medium	11.63948, 78.12081	30 KM/Hr
19	Turn	High	11.64953, 78.40233	15 KM/Hr
20	Turn	Medium	11.64476, 78.40198	30 KM/Hr
21	Turn	High	11.64429, 78.40220	15 KM/Hr
22	Turn	High	11.64407, 78.40406	15 KM/Hr
23	Turn	Medium	11.62347, 78.40744	30 KM/Hr
24	Turn	High	11.61234, 78.41126	15 KM/Hr
25	Turn	Medium	11.60904, 78.41789	30 KM/Hr
26	Turn	Medium	11.59837, 78.43873	30 KM/Hr
27	Turn	Medium	11.59152, 78.44336	30 KM/Hr
28	Turn	High	11.58768, 78.44445	15 KM/Hr
29	Blind Spot	Blind Spot	11.58332, 78.44386	10 KM/Hr
30	Turn	Medium	11.57494, 78.49402	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
31	Blind Spot	Blind Spot	11.57100, 78.49943	10 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Kathir Eye Hospital	11.4740976, 77.8681591	30 km/h	Medium
1	police	Police Quarters	11.4741243, 77.8673495	30 km/h	Medium
2	hospital	Thilagam Hospital	11.4753573, 77.8685829	30 km/h	Medium
3	clinic	Meyyazhagan Clinic	11.4752254, 77.8685671	30 km/h	Medium
4	hospital	Dr. Jaganathan Hospital	11.4752717, 77.8709312	30 km/h	Medium
5	hospital	Sankari Hospital	11.4777496, 77.873892	30 km/h	Medium
7	clinic	Dr. Dhanasekar Clinic	11.4793113, 77.8829247	30 km/h	Medium
8	hospital	Sri Vellaiyan Hospital	11.4798876, 77.8831394	30 km/h	Medium
10	hospital	C. N. Hospital	11.55724, 78.0096	30 km/h	Medium
11	hospital	Vinayaka Mission's Homeopathic Medical College & Hospital	11.583617, 78.059145	30 km/h	Medium
12	hospital	Vinayaka mission Kirubananadah variyar medical college hospital	11.584984, 78.0618417	30 km/h	Medium
13	hospital	vims	11.5854239, 78.0639188	30 km/h	Medium
16	hospital	Vinayaka Mission’s Kirupananda Variyar Medical College	11.5977669, 78.0850992	30 km/h	Medium
20	hospital	Priyam Speciality Hospital	11.63446, 78.124388	30 km/h	Medium
22	hospital	Universal Cancer Hospital	11.6336739, 78.1267767	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
23	hospital	AVM Hospital	11.6264369, 78.1394204	30 km/h	Medium
24	hospital	Dharan Hospital	11.6252302, 78.1451753	30 km/h	Medium
25	hospital	Udhayam Hospital	11.625819, 78.145958	30 km/h	Medium
26	pharmacy	மருந்தகம்	11.6251501, 78.1470098	30 km/h	Medium
27	hospital	Rps Hospital	11.6260528, 78.1474921	30 km/h	Medium
28	hospital	மருத்துவமனை நிலம்	11.625344, 78.146058	30 km/h	Medium
29	hospital	Akitha Hospital	11.6256027, 78.1489875	30 km/h	Medium
30	hospital	Deepam Speciality Hospital	11.6269734, 78.1476357	30 km/h	Medium
31	hospital	Manima Clinic	11.623248, 78.1465721	30 km/h	Medium
35	hospital	Jaaidev Hospital	11.6655205, 78.2634736	30 km/h	Medium
37	clinic	Dr. Saravanakumar Clinic	11.5842002, 78.4441223	30 km/h	Medium
38	hospital	Government Hospital	11.572472, 78.497407	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
6	marketplace	Sunday Market	11.4796502, 77.8708458	30 km/h	Medium
14	university	Vinayaka Missions Vinayaka Missions Research Foundation	11.5974056, 78.0843617	30 km/h	Medium
15	school	Vinayaka Missions Annapoorani Matric School	11.5971662, 78.0839876	30 km/h	Medium
17	school	Dist Education Training	11.6133799, 78.1022409	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
18	school	DIET - Salem	11.6144158, 78.1025078	30 km/h	Medium
19	school	Diamond Rays Matric HR Sec School	11.6385629, 78.1206259	30 km/h	Medium
21	school	Sowdeshwari College	11.6330643, 78.1276021	30 km/h	Medium
32	school	Ganesh College	11.6566575, 78.1991121	30 km/h	Medium
33	school	Thangam ZEE public school	11.6653353, 78.251641	30 km/h	Medium
34	college	AVS College od Arts and Science, Salem.	11.6642672, 78.2584134	30 km/h	Medium
36	college	Mahendra Engineering College	11.6672537, 78.2641303	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.63303, 78.12713



Risk Type: U-Turn

Risk Level: High

Speed Limit: 10 KM/Hr

Coordinates: 11.5709952, 78.4994313



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44029, 77.87544

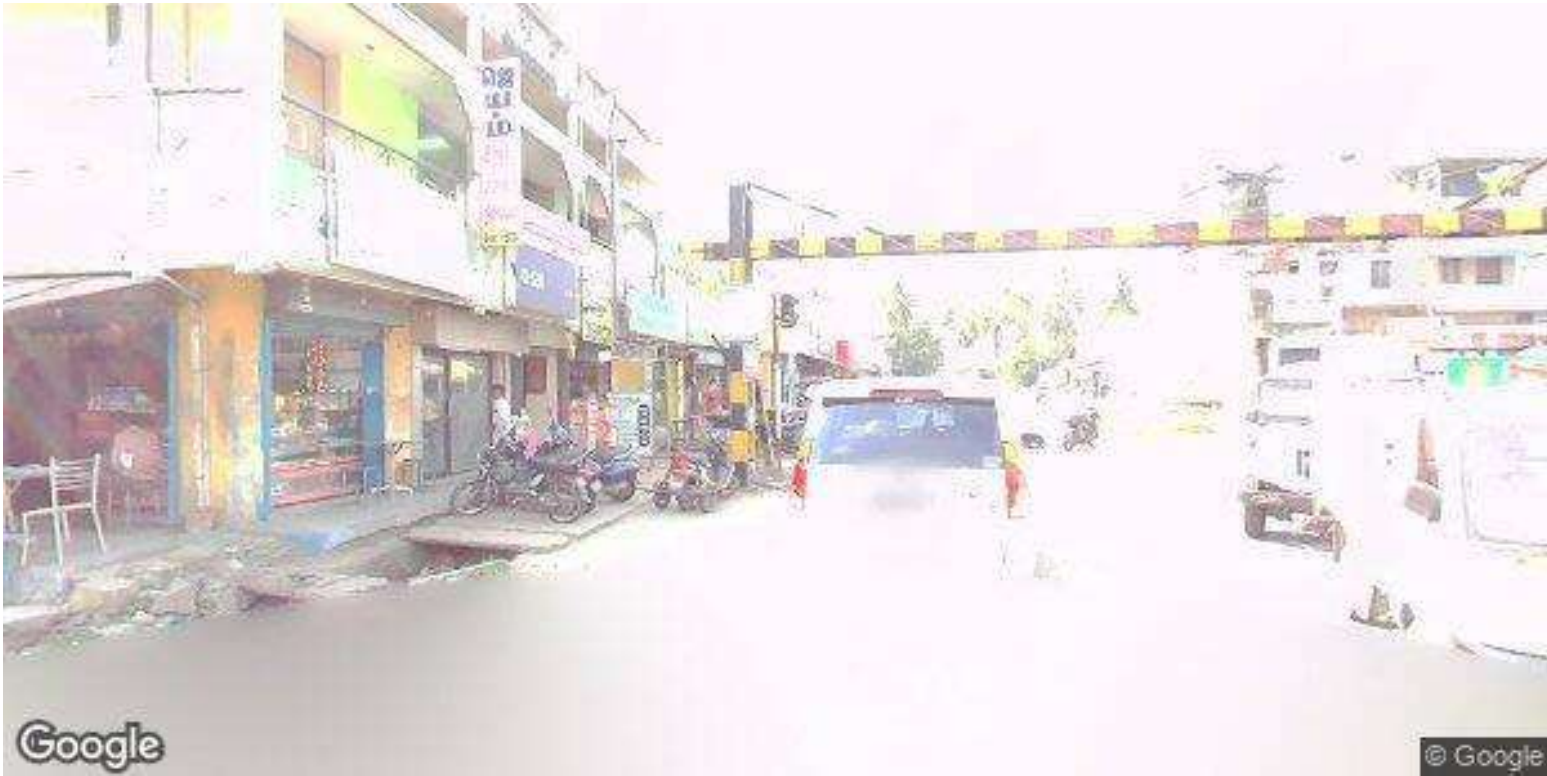


Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.44900, 77.87406



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45000, 77.87213



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45028, 77.87202



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45309, 77.87325



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.46400, 77.87323



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46765, 77.87037



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.47403, 77.86841



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49149, 77.89601



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.49224, 77.89601



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.64172, 78.11817



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64216, 78.11875



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64195, 78.12129



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64162, 78.12160



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.64120, 78.12168



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.63948, 78.12081



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.64953, 78.40233



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64476, 78.40198



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.64429, 78.40220



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.64407, 78.40406



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.62347, 78.40744



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.61234, 78.41126



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.60904, 78.41789



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.59837, 78.43873



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.59152, 78.44336



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.58768, 78.44445



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.58332, 78.44386



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.57494, 78.49402



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.57100, 78.49943

